

Math 421

Monday, November 3

Announcements

- Homework 8 Due Friday
- Exam 2 – 11/12 5:45-7:15 pm
 - Email me & fill out the form if you have a conflict and need to take the makeup.
 - Sign up for McBurney exam if applicable.

Activity – Critique the Proof

Theorem. $\lim_{x \rightarrow 1} x^2 + x + 1 = 3$.

False Proof. Fix $\epsilon > 0$. Set $\delta = \frac{\epsilon}{|x+2|}$. Assume $0 < |x - 1| < \delta$.

$$\begin{aligned} \text{Then } |(x^2 + x + 1) - 3| &= |x^2 + x - 2| = |(x - 1)(x + 2)| \\ &= |x - 1||x + 2| < \delta \cdot |x + 2| = \frac{\epsilon}{|x+2|} \cdot |x + 2| = \epsilon. \end{aligned}$$

Since $\epsilon > 0$ was arbitrary, $\lim_{x \rightarrow 1} x^2 + x + 1 = 3$.

Example – $\lim_{x \rightarrow 1} x^2 + x + 1 = 3$

Discussion.

Algorithm for quadratic polynomials:

1. Factor out $|x - a|$ from $|f(x) - L|$
2. Deal with the extra factor
 - a) Write in terms of $x - a$:
 - b) Use triangle inequality:
 - c) Assume a bound of 1:
3. Set $\delta = \min\{1, \epsilon/\text{answer in c)}\}$

Example – $\lim_{x \rightarrow 1} x^2 + x + 1 = 3$

Proof.

Activity – Fill in the Blanks

Thm. $\lim_{x \rightarrow 3} x^2 - x = 6.$

Proof. Fix $\epsilon > 0$. Set $\delta = \min\{1, \frac{\epsilon}{2}\}$. Assume $0 < |x - 3| < \delta$.

Then $|x^2 - x - 6| =$ _____

 _____ $< \epsilon$.

Since $\epsilon > 0$ was arbitrary, $\lim_{x \rightarrow 3} x^2 - x = 6$.

Example – $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0$

Discussion.

Proof.

Functional Limits – via Sequences

Def. Let $f: S \rightarrow \mathbb{R}$ and let a be the limit of some sequence in S .

Then $\lim_{x \rightarrow a} f(x) = L$ if for every sequence (x_n) in S satisfying $x_n \neq a$ for all $n \in \mathbb{N}$ and $\lim x_n = a$, we have $\lim f(x_n) = L$.

Example: Let $f(x) = x^2$, $\lim_{x \rightarrow 0} x^2 = 0$.

Activity

Prove carefully using sequences, and their algebraic limit laws, that

$$\lim_{x \rightarrow 3} x^2 - x = 6.$$

Proof.

Proof of Equivalence

$(\epsilon - \delta \Rightarrow \text{Sequences})$:

Proof of Equivalence

$(\epsilon - \delta \Leftarrow \text{Sequences})$:

Algebraic Limit Laws

Theorem. Let f and g be functions defined on a domain S and let a be the limit of some sequence in S . If the limits $L = \lim_{x \rightarrow a} f(x)$ and $M = \lim_{x \rightarrow a} g(x)$ exist and are finite. Then

- i. $\lim_{x \rightarrow a} kf(x)$ exists and equals kL for all $k \in \mathbb{R}$.
- ii. $\lim_{x \rightarrow a} (f + g)(x)$ exists and equals $L + M$.
- iii. $\lim_{x \rightarrow a} (fg)(x)$ exists and equals LM .
- iv. $\lim_{x \rightarrow a} (f/g)(x)$ exists and equals L/M provided $M \neq 0$ and $g(x) \neq 0$ for $x \in \text{dom}(g)$.

Activity – Divide & Conquer the proofs of i, ii, and iii.